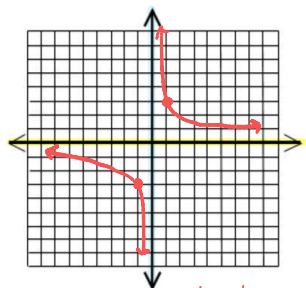


$$1. f(x) = \frac{3}{x}$$

VA at $x=0$
HA at $y=0$ (end behavior)



y-int None $f(0) = \frac{3}{0}$ undefined!

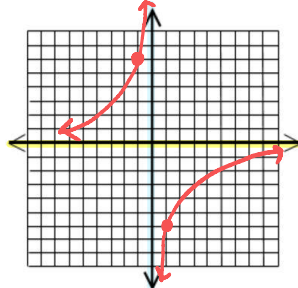
x-int none $\frac{3}{x} = 0$
 $3 = 0$ false!

to help graph
 $f(1) = 3$
 $f(-1) = -3$

Domain: $\mathbb{R} \text{ ex. } 0$
Range: $\mathbb{R} \text{ ex. } 0$

$$2. g(x) = \frac{-12}{x}$$

VA at $x=0$
HA at $y=0$ (end behavior)



y-int None $f(0) = \frac{-12}{0}$ undefined!

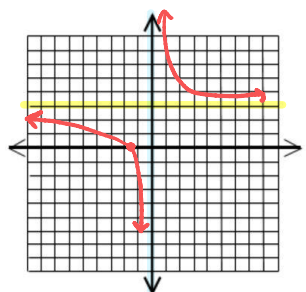
x-int none $\frac{-12}{x} = 0$
 $-12 = 0$ false!

$f(1) = -12$
 $f(-1) = 12$

Domain: $\mathbb{R} \text{ ex. } 0$
Range: $\mathbb{R} \text{ ex. } 0$

$$3. g(x) = \frac{4}{x} + 3$$

VA at $x=0$
HA at $y=3$ (end behavior)



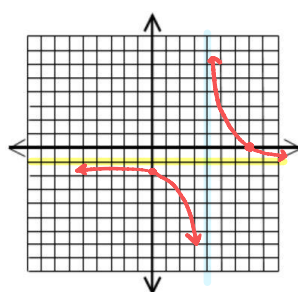
y-int none $f(0) = \frac{4}{0} + 3$ undefined!

x-int $\frac{4}{x} + 3 = 0$
 $\frac{4}{x} = -3$ $4 = -3x$
 $x = -\frac{4}{3}$

Domain: $\mathbb{R} \text{ ex. } 0$
Range: $\mathbb{R} \text{ ex. } 3$

$$4. y = \frac{3}{x-4} - 1$$

VA at $x=4$
HA at $y=-1$ (end behavior)



y-int $f(0) = \frac{3}{0-4} - 1$
 $= \frac{3}{-4} - \frac{4}{4} = \frac{-7}{4}$

x-int $\frac{3}{x-4} - 1 = 0$ $\frac{3}{x-4} = 1$ $x-4=3$ $x=7$

Domain: $\mathbb{R} \text{ ex. } 4$
Range: $\mathbb{R} \text{ ex. } -1$

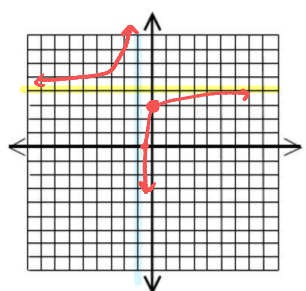
$$5. g(x) = \frac{4x+3}{x+1}$$

Synthetic division

$$\begin{array}{r} -1 \overline{) 4 \ 3} \\ \underline{4 \ -4} \\ 4 \ -1 \end{array}$$

$$4 + \frac{-1}{x+1}$$

$$f(x) = -\frac{1}{x+1} + 4$$



VA at $x=-1$ Domain: $\mathbb{R} \text{ ex. } -1$
HA at $y=4$ Range: $\mathbb{R} \text{ ex. } 4$

x-intercept $4x+3=0$
 $4x=-3$
 $x = -\frac{3}{4}$

y-int $g(0) = \frac{3}{1}$

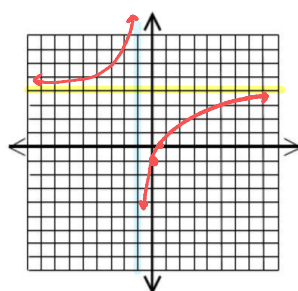
$(-\frac{3}{4}, 0)$ $(0, 3)$

$$6. h(x) = \frac{4x-1}{x+1}$$

$$\begin{array}{r} -1 \overline{) 4 \ -1} \\ \underline{4 \ -4} \\ 4 \ -5 \end{array}$$

$$4 + \frac{-5}{x+1} \rightarrow h(x) = \frac{-5}{x+1} + 4$$

VA at $x=-1$ Domain: $\mathbb{R} \text{ ex. } -1$
HA at $y=4$ Range: $\mathbb{R} \text{ ex. } 4$

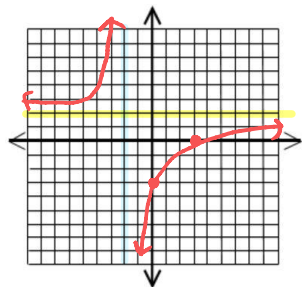


x-int $4x-1=0$
 $4x=1$
 $x = \frac{1}{4}$
 $(\frac{1}{4}, 0)$

y-int $h(0) = \frac{-1}{1}$
 $= -1$
 $(0, -1)$

$$7. f(x) = \frac{2x-6}{x+2}$$

$$\begin{array}{r} -2 \overline{) 2 \quad -6} \\ \underline{2 \quad -10} \end{array}$$



$$2 + \frac{-10}{x+2} \rightarrow f(x) = \frac{-10}{x+2} + 2$$

VA: $x = -2$ Domain: $\mathbb{R}$ except -2

HA: $y = 2$ Range: $\mathbb{R}$ ex. 2

x-int:

$$2x - 6 = 0$$

$$2x = 6$$

$$x = 3$$

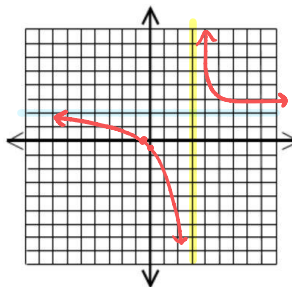
$$(3, 0)$$

y-int

$$f(0) = \frac{-6}{2} = -3 \quad (0, -3)$$

$$8. y = \frac{2x+1}{x-3}$$

$$\begin{array}{r} 3 \overline{) 2 \quad 1} \\ \underline{2 \quad 7} \end{array}$$



$$y = \frac{7}{x-3} + 2$$

VA at $x = 3$

HA at $y = 2$

Domain: $\mathbb{R}$ ex. 3

Range: $\mathbb{R}$ ex. 2

x-intercept

$$2x + 1 = 0$$

$$2x = -1$$

$$x = -\frac{1}{2}$$

$$(-\frac{1}{2}, 0)$$

y-intercept

$$y(0) = \frac{1}{-3}$$

$$(0, -\frac{1}{3})$$

$$9. \frac{3x^2-5x-2}{x^2-4} = \frac{3x+1}{x+2}$$

$$\begin{array}{l} (3x^2-6x)(x-2) \\ 3x(x-2)+1(x-2) \\ (3x+1)(x-2) \end{array}$$

$$\frac{(3x+1)(x-2)}{(x+2)(x-2)}$$

$$\frac{3x+1}{x+2}$$

$$10. \frac{x^2-25}{x^3-125} = \frac{x+5}{x^2+5x+25}$$

$$\text{comp} \rightarrow x^3-125 \quad x^2+5x+25$$

$$\frac{(x+5)(x-5)}{(x-5)(x^2+5x+25)} = \frac{x+5}{x^2+5x+25}$$

$$11. \frac{3x^2-12}{5x-10} \cdot \frac{1}{2x+4} = \frac{3}{10}$$

$$\frac{3(x^2-4)}{5(x-2)} \cdot \frac{1}{2(x+2)}$$

$$\frac{3(x+2)(x-2)}{5(x-2)} \cdot \frac{1}{2(x+2)}$$

$$\frac{3}{5 \cdot 2} = \frac{3}{10}$$

$$12. \frac{21x^{10}y^5}{5x^2} \cdot \frac{x^3}{35y^4} = \frac{3x^{11}y}{25}$$

$$21x^8y^5 \cdot \frac{x^3}{35y^4} = \frac{21x^{11}y}{35}$$

$$\frac{3x^{11}y}{25}$$

$$13. (x^2 + 10x - 24) \div \frac{x^2 - 144}{3x - 36} = 3x - 6$$

$$x^2 + 10x - 24 \cdot \frac{3x - 36}{x^2 - 144}$$

$$(x-2)(x+12) \cdot \frac{3(x-12)}{(x+12)(x-12)}$$

$$3(x-2) \text{ or } 3x-6$$

$$14. \frac{x^3 - 8}{64x} \div \frac{x^2 - x - 2}{16x^2} = \frac{x(x^2 + 2x + 4)}{4(x+1)}$$

$$\frac{x^3 - 8}{64x} \cdot \frac{16x^2}{x^2 - x - 2}$$

$$\frac{x^2 + 2x + 4}{4} \cdot \frac{x}{x+1} = \frac{x(x^2 + 2x + 4)}{4(x+1)}$$

$$15. \frac{\frac{x+6}{5} + \frac{1}{x}}{\frac{5}{x} + \frac{25}{x^2}} = \frac{x(x+1)}{25}$$

$$\frac{x}{x} \cdot \frac{x+6}{5} + \frac{1}{x} \cdot \frac{5}{5}$$

$$\frac{x}{x} \cdot \frac{5}{x} + \frac{25}{x^2}$$

$$\frac{x^2 + 6x}{5x} + \frac{5}{5x}$$

$$\frac{5x}{x^2} + \frac{25}{x^2}$$

$$\frac{x^2 + 6x + 5}{5x}$$

$$\frac{5x + 25}{x^2}$$

$$\frac{x^2 + 6x + 5}{5x} \xrightarrow{\text{keep}} \frac{5x + 25}{x^2} \xrightarrow{\text{flip}}$$

$$\frac{x^2 + 6x + 5}{5x} \cdot \frac{x^2}{5x + 25} = \frac{(x+5)(x+1)}{5x} \cdot \frac{x^2}{5(x+5)}$$

$$\frac{x(x+1)}{25}$$

$$16. \frac{\frac{1}{2x-5} - \frac{7}{8x-20}}{\frac{x}{2x-5}} = \frac{-3}{4x}$$

$$\frac{4}{4} \cdot \frac{1}{2x-5} - \frac{7}{4(2x-5)}$$

$$\frac{4}{4(2x-5)} - \frac{7}{4(2x-5)} = \frac{-3}{4(2x-5)}$$

$$\frac{\frac{-3}{4(2x-5)}}{\frac{x}{2x-5}}$$

$$\frac{-3}{4(2x-5)} \cdot \frac{2x-5}{x} = \frac{-3}{4x}$$

$$17. \frac{\frac{1}{3x^2-3}}{\frac{5}{x+1} - \frac{x+4}{x^2-3x-4}} = \frac{x-4}{12(x-1)(x-6)}$$

$$\frac{1}{3(x^2-1)} = \frac{1}{3(x+1)(x-1)}$$

$$\frac{(x-4)5}{(x-4)(x+1)} - \frac{x+4}{(x-4)(x+1)}$$

$$\frac{5(x-4) - (x+4)}{(x-4)(x+1)} = \frac{5x-20-x-4}{(x-4)(x+1)} = \frac{4x-24}{(x-4)(x+1)}$$

$$\frac{1}{3(x+1)(x-1)} \xrightarrow{\text{keep}} \frac{4x-24}{(x-4)(x+1)} \xrightarrow{\text{flip}}$$

$$\frac{1}{3(x+1)(x-1)} \cdot \frac{(x-4)(x+1)}{4x-24} = \frac{x-4}{3(x-1)(4x-24)} = \frac{x-4}{12(x-1)(x-6)}$$

$$18. \frac{x}{x^2-10} = \frac{3}{2x+1} \quad x = -5, x = 6$$

$$x(2x+1) = 3(x^2-10)$$

$$2x^2 + x = 3x^2 - 30$$

$$0 = x^2 - x - 30$$

$$0 = (x-6)(x+5)$$

$$x = 6, x = -5$$

$$19. \frac{3}{x} - \frac{2}{x+1} = \frac{4}{x} \quad x = -\frac{1}{3}$$

$$\left[\frac{3}{x} - \frac{2}{x+1} = \frac{4}{x} \right] \times (x+1)$$

$$\frac{3}{x} \times (x+1) - \frac{2}{x+1} \times (x+1) = \frac{4}{x} \times (x+1)$$

$$x \neq 0, -1$$

$$3(x+1) - 2x = 4(x+1)$$

$$3x+3-2x=4x+4$$

$$x+3=4x+4$$

$$-1=3x$$

$$x = -\frac{1}{3}$$

$$20. \frac{2}{2x+5} + \frac{3}{2x-5} = \frac{5x+5}{4x^2-25} \quad x = 0$$

$$x \neq -\frac{5}{2}, \frac{5}{2}$$

$$(2x-5)(2x+5) \left[\frac{2}{2x+5} + \frac{3}{2x-5} = \frac{5x+5}{(2x-5)(2x+5)} \right] (2x-5)(2x+5)$$

$$2(2x-5) + 3(2x+5) = 5x+5$$

$$4x-10+6x+15=5x+5$$

$$10x+5=5x+5$$

$$5x=0$$

$$x=0$$

$$21. \frac{1}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6} \quad \text{No solution}$$

$$x \neq 2, -3$$

$$(x+3)(x-2) \left[\frac{1}{x-2} + \frac{1}{x+3} = \frac{5}{(x+3)(x-2)} \right] (x+3)(x-2)$$

$$x+3+x-2=5$$

$$2x+1=5$$

$$2x=4$$

$$x=2 \text{ extraneous}$$

No solution

$$22. \frac{x}{5x+5} = \frac{1}{x+2} + \frac{1}{x^2+3x+2} \quad x = 5$$

$$x \neq -2, -1$$

$$5(x+1)(x+2) \left[\frac{x}{5(x+1)} = \frac{1}{x+2} + \frac{1}{(x+1)(x+2)} \right] 5(x+1)(x+2)$$

$$x(x+2) = 5(x+1) + 5$$

$$x^2 + 2x = 5x + 5 + 5$$

$$x^2 + 2x = 5x + 10$$

$$x^2 - 3x - 10 = 0$$

$$(x-5)(x+2) = 0$$

$$x=5, x=-2 \text{ extraneous}$$

$$23. \frac{3}{x^2} = \frac{x-4}{3x^2} + \frac{2}{3x^2} \quad x = 11$$

$$\left[\frac{9}{3x^2} = \frac{x-4}{3x^2} + \frac{2}{3x^2} \right] 3x^2$$

$$x \neq 0$$

$$9 = x-4+2$$

$$9 = x-2$$

$$x = 11$$

$$24. \frac{1}{x} + \frac{3x+12}{x^2-5x} = \frac{7x-56}{x^2-5x} \quad x = 21$$

$$\frac{x-5}{x-5} \cdot \frac{1}{x} + \frac{3x+12}{x(x-5)} = \frac{7x-56}{x(x-5)} \quad x \neq 0, 5$$

$$x-5+3x+12=7x-56$$

$$4x+7=7x-56$$

$$63=3x$$

$$x=21$$

$$25. \frac{2}{x^2-2x-3} = \frac{3x}{x+1} + \frac{1}{2x-6} \quad x = -\frac{1}{6}$$

$$\left[\frac{2}{(x-3)(x+1)} = \frac{3x}{x+1} + \frac{1}{2(x-3)} \right] 2(x-3)(x+1)$$

$$4 = 6x(x-3) + x+1$$

$$4 = 6x^2 - 18x + x + 1$$

$$4 = 6x^2 - 17x + 1$$

$$0 = 6x^2 - 17x - 3$$

$$(6x+1)(x-3) = 0$$

$$6x^2 - 18x + x - 3$$

$$6x(x-3) + 1(x-3)$$

$$(6x+1)(x-3)$$

$$x = -\frac{1}{6}, x=3 \text{ extraneous}$$

26. You are creating statues made of cement. The mold costs \$300. The materials for each statue costs \$22.

- a. Estimate how many statues must be made for the average cost per statue to fall below \$30.

$$\frac{300 + 22x}{x} = \frac{30}{1}$$

$$\begin{aligned} 300 + 22x &= 30x \\ 300 &= 8x \\ x &= 37.5 \end{aligned}$$

38 statues

- b. What happens to the average cost as more statues are created?

As more statues are created,
the average cost approaches \$22

27. The senior class is sponsoring a dinner. The cost of catering the dinner is \$9.95 per person plus an \$18 delivery charge.

- a. Write a model that gives the average cost per person.

$$A(x) = \frac{18 + 9.95x}{x}$$

- b. Estimate the number of people needed to lower the cost to \$11 per person.

$$\frac{11}{1} = \frac{18 + 9.95x}{x}$$

$$\begin{aligned} 11x &= 18 + 9.95x \\ 1.05x &= 18 \\ x &= 17.14 \end{aligned}$$

18 people

- c. Find and interpret the horizontal asymptote of your model.

HA at $y = 9.95$

As the number of people gets bigger,
the average cost per dinner gets closer to \$9.95

- d. Find and interpret the vertical asymptote of your model.

VA at $x = 0$

$A(x)$ is undefined at $x = 0$ because
there can be no average cost for 0 dinners

28. The average cost, in dollars, per unit $C(x)$ to produce x units of plywood is given by $C(x) = \frac{300}{x+10}$.

- a. What is the cost per unit when 590 units are produced?

$$\begin{aligned} \underbrace{\quad}_{x=590} \quad C(590) &= \frac{300}{590+10} = \frac{300}{600} = \frac{1}{2} \end{aligned}$$

When 590 units are produced, the average cost is \$0.50

- b. If the cost per unit is \$1.50, how many units have been produced?

$$1.5 = \frac{300}{x+10}$$

$$\begin{aligned} 1.5(x+10) &= 300 \\ 1.5x + 15 &= 300 \\ 1.5x &= 285 \\ x &= 190 \end{aligned}$$

190 units

29. A rare species of insect was discovered in the rainforest of Costa Rica. Environmentalists transplant the insect into a protected area. The population of the insect t months after being transplanted is

$$P(t) = \frac{45(1+0.6t)}{(3+0.02t)}$$

- a. What was the population when $t = 0$?

$$P(0) = \frac{45(1+0.6(0))}{3+0.02(0)} = \frac{45}{3} = 15 \quad \text{15 insects}$$

- b. What will the population be after 10 years?

$$P(10) = \frac{45(1+0.6(10))}{3+0.02(10)} = \frac{315}{3.2} = 98.4375 \quad \text{98 insects}$$

- c. When will there be 549 insects?

$$\frac{549}{1} = \frac{45(1+0.6t)}{3+0.02t}$$

$$\begin{aligned} 549(3+0.02t) &= 45(1+0.6t) \\ 1647 + 10.98t &= 45 + 27t \\ 1602 &= 36.72t \\ t &= 43.63 \end{aligned}$$

After 44 years

30. For a fundraising project, your math club is publishing a graphic art calendar. The cost of the digital images and the permission to use them is \$850. In addition to these "one-time" charges, the unit cost of printing each calendar is \$3.25.

- a. Write a model that gives the average cost per calendar as a function of the number of calendars printed.

$$C(x) = \frac{850 + 3.25x}{x}$$

- b. Estimate the number of calendars you need to print before the average cost drops to \$5 per calendar.

$$5 = \frac{850 + 3.25x}{x}$$

$$\begin{aligned} 5x &= 850 + 3.25x \\ 1.75x &= 850 \end{aligned}$$

$$x = 485.71$$

At least 486 calendars

- c. Describe what happens to the average cost as the number of calendars printed increases.

As the number of calendars increases, the cost per calendar gets closer to \$3.25

- d. Suppose you decide to generate your own designs on a computer to save money. The cost for the software (a "one-time" cost) is \$125. Write a model that gives the average cost per calendar as a function of the number of calendars printed. Estimate the number of calendars you need to print before the average cost drops to \$5 per calendar.

$$5 = \frac{125 + 3.25x}{x}$$

$$\begin{aligned} 5x &= 125 + 3.25x \\ 1.75x &= 125 \end{aligned}$$

$$x = 71.4$$

At least 72 calendars